

Can Chance Explain the Mean Difference?

Topic 7.1 · TODAY'S PROBLEM

Suppose a researcher recruits 50 student volunteers to test whether a planning prompt lowers mean logic-task time, then randomly assigns 25 to the prompt and 25 to standard directions. Their means are 14.2 and 17.8 minutes, so prompt minus standard is -3.6 . Under no effect, 18 of 2,000 rerandomized differences are -3.6 or lower.

Nia: “A result this favorable is rare under no effect, supporting a causal decrease for students like these volunteers.”

Owen: “The 0.009 is the probability of no effect, so there is a 99.1% chance the prompt works for every district student.”

Experiment results

Group	n	Mean (min)
Prompt	25	14.2
Standard	25	17.8

FIRST TAKE (5 min, on your sheet)

Whose use of the rerandomization result is better supported, Nia's or Owen's? Cite -3.6 or 18/2,000.

- DOK 1** (a) State the directional claim, the observed evidence statistic, and its expected value under no effect.
- DOK 2** (b) Describe one rerandomization trial, compute the tail proportion, and interpret it under no effect.
- DOK 3** (c) Decide whose conclusion is warranted using 0, -3.6 , and 18 of 2,000. Distinguish the two directions of conditional probability, bound causation and generalization from the design, identify the remaining inference error, and explain the rejected statement's consequence.

Video + follow-along: u7_lesson1_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

